

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

AI for Medical Image Analysis [CSE-4011]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With appropriate examples, compare and contrast the various segmentation strategies (5)
 - A)
 - B) Relate how space and time continuity contribute to the task of image segmentation (3)
 - C) Predict whether the following statement is true or false and justify your answer. "Segmentation in a noisy environment is difficult than it seems". (2)
- 2) Investigate the operational mechanics of the Scale Invariant Feature Transform algorithm, elaborating on its underlying principles. (5)
 - A)
 - B) Demonstrate the significance of Laplacian of Gaussian in Blob detection task with example. (3)
 - C) Compare and contrast multiresolution edge detection and multi-resolution edge tracking with examples. (2)
- 3) Compare and contrast TensorFlow, Caffe, and Theano deep learning frameworks in terms of their suitability for medical image analysis tasks. What are the key features, advantages, and limitations of each framework, particularly in handling large-scale medical image datasets? (5)
 - A)
 - B) Relate the capabilities of MATLAB 2020b for medical image processing and analysis. (3)
 - C) Consider the following grayscale image of size 5x4.
Apply and design $C_{boxcar,3}$ filter to the given image. (2)

45	100	105	20	20
60	60	50	120	250
250	250	100	200	220
10	20	20	40	40

- 4) A radiology department in a hospital is overwhelmed with the manual segmentation of organs from CT scans for treatment planning in radiation therapy. They seek to implement an automated segmentation solution to streamline the process and improve efficiency. With the provided scenario in mind, determine a detailed strategy for developing and deploying an automated organ segmentation algorithm for CT scans in the context of radiation therapy treatment planning. (5)
- A) For the scenario given below, determine the preprocessing steps necessary to standardize image intensities, correct imaging artifacts, and normalize spatial dimensions across the dataset.
- B) A radiology department in a hospital is overwhelmed with the manual segmentation of organs from CT scans for treatment planning in radiation therapy. They seek to implement an automated segmentation solution to streamline the process and improve efficiency. (3)
- C) For the scenario given below, determine how would you validate the developed system.
- A radiology department in a hospital is overwhelmed with the manual segmentation of organs from CT scans for treatment planning in radiation therapy. They seek to implement an automated segmentation solution to streamline the process and improve efficiency. (2)
- 5) Contrast a difference between analysis tasks for a clinical study compared to image analysis in computer-assisted surgery and name at least two analysis tasks that could arise when using images for diagnosis support. Infer the main problem that arises when validating an analysis method. (5)
- A)
- B) Contrast the difference between an HIS and an RIS. Why is it useful to have several different information systems in a hospital? (3)
- C) Explain the structure of a DICOM file. What are the general components and how is the structure of a tagged information entity in the file? (2)

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